SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 COURSE CODE: Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Concept Mapping

Code of Conduct:

I VAIBHAYA LAKSHMI.K (192511310) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vaibhaya. K

Annexure A

Concept Mapping

Question 1
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

Question 2
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

Question 3
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases. **(BL2)**

Question 4
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

Question 5
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

3)

Operating System Types

Batch OS

- Groups jobs & executes them in batches

⇒ Advantage:
- Efficient for repetitive jobs.
- High throughput

⇒ Limitations:
- Long response time.

Time-Sharing OS

- Allocates CPU time slices to multiple users or processes.

⇒ Advantage:
- Interactive & quick response.
- Better resource utilisation.

⇒ Limitations:
- Frequent switching overhead.

Distributed OS

- Coordinates multiple networked computers & resources.

⇒ Advantage:
- Resource sharing.
- Scalability & reliability.

⇒ Limitations:
- Depends on network communication.

Real-Time OS

- Executes tasks within specified timing constraints.

⇒ Advantage:
- High reliability
- Predictable & timely response.

⇒ Limitations
- Limited flexibility.

Similarities

- Manage computer resources & execute jobs
- Provide interface b/w application & hardware.
- Require scheduling & management of resource.

Dynamic Concept Map

